

1. THE KOHLER–JOBIN INEQUALITY AND THE TRANSFER TO FABER–KRAHN

This section is logically self-contained. In Part 1 we reconstruct from scratch, by the Kohler–Jobin rearrangement technique, the spectral shape inequality

$$T(\Omega) \lambda_1(\Omega)^{(n+2)/2} \geq T(B) \lambda_1(B)^{(n+2)/2} \quad (B \text{ any ball}),$$

with equality only for balls. In Part 2 we feed this inequality, together with the global sharp Saint–Venant stability estimate proved elsewhere in the manuscript, into a short and fully explicit computation that yields sharp quantitative Faber–Krahn stability, $\delta_{\text{FK}}(\Omega) \geq c_{\text{FK}}(n) \mathcal{A}(\Omega)^2$.

The technique, due to Kohler–Jobin [?, ?, ?], is a spherical rearrangement that, unlike Schwarz symmetrisation, preserves not the *volume* of the superlevel sets but a *torsional* functional of them. We follow the modern exposition of Brasco [?, §§3–5, Thm. 1.1], specialised throughout to the linear case (the Laplacian, $p = 2$, and the Dirichlet energy quotient).

1.1. Notation and scaling conventions. Throughout, $n \geq 2$ and $\Omega \subset \mathbb{R}^n$ is open with finite measure. We use:

- $u_\Omega \in H_0^1(\Omega)$, the *torsion function*, the unique weak solution of $-\Delta u_\Omega = 1$ in Ω , $u_\Omega = 0$ on $\partial\Omega$;
- $T(\Omega) = \int_\Omega u_\Omega \, dx$, the *torsional rigidity*, and $E(\Omega) = -T(\Omega)/2$;
- $\lambda_1(\Omega)$, the first Dirichlet eigenvalue,

$$(1) \quad \lambda_1(\Omega) = \min_{0 \neq v \in H_0^1(\Omega)} \frac{\int_\Omega |\nabla v|^2 \, dx}{\int_\Omega v^2 \, dx},$$

attained by a first eigenfunction $u > 0$ solving $-\Delta u = \lambda_1(\Omega)u$;

- B^* , the ball centred at 0 with $|B^*| = |\Omega|$; B_r the open ball of radius r centred at 0; B_1 the unit ball; $\omega_n = |B_1|$;
- $L_n = \lambda_1(B_1) = j_{n/2-1,1}^2$ and $T_n = T(B_1) = \omega_n/(n(n+2))$;
- $\mathcal{A}(\Omega) = \inf_{x_0 \in \mathbb{R}^n} |\Omega \triangle (x_0 + B^*)| / |B^*| \in [0, 2)$, the Fraenkel asymmetry;
- $\delta_{\text{FK}}(\Omega) = (|\Omega|/\omega_n)^{2/n} (\lambda_1(\Omega) - \lambda_1(B^*))$, the scale-invariant Faber–Krahn deficit.

The torsion function of a ball is $u_{B_r}(x) = (r^2 - |x|^2)/(2n)$, whence

$$(2) \quad T(B_r) = \frac{\omega_n}{n(n+2)} r^{n+2} = T_n r^{n+2}.$$

Under dilation $T(t\Omega) = t^{n+2}T(\Omega)$ and $\lambda_1(t\Omega) = t^{-2}\lambda_1(\Omega)$, so

$$(3) \quad \Psi(\Omega) := T(\Omega) \lambda_1(\Omega)^{(n+2)/2}$$

is scale invariant; in particular $\Psi(B_r) = \Psi(B_1) = T_n L_n^{(n+2)/2}$ for every ball. We abbreviate $\beta := (n+2)/2$.

1.2. Part 1: the Kohler–Jobin inequality, proved from scratch.

Theorem 1.1 (Kohler–Jobin). *For every open $\Omega \subset \mathbb{R}^n$ of finite measure with $\lambda_1(\Omega) < \infty$ and every ball $B \subset \mathbb{R}^n$,*

$$(4) \quad T(\Omega) \lambda_1(\Omega)^\beta \geq T(B) \lambda_1(B)^\beta, \quad \beta = \frac{n+2}{2}.$$

Equivalently $\Psi(\Omega) \geq \Psi(B_1) = T_n L_n^\beta$, with equality if and only if Ω is a ball (up to translation and a Lebesgue-null modification).

Strategy. Both sides of (??) have the same scaling and $\Psi(B) = \Psi(B_1)$ for every ball; so it suffices to compare Ω with a single, well-chosen ball. The idea is to rearrange the first eigenfunction u of Ω into a radial decreasing function u^* on a ball B in such a way that

- (A) the Dirichlet energy is *exactly preserved*, $\int_B |\nabla u^*|^2 = \int_\Omega |\nabla u|^2$, and
- (B) the L^2 norm *does not decrease*, $\int_B (u^*)^2 \geq \int_\Omega u^2$.

Granting ??-??, the Rayleigh principle (??) on B gives

$$(5) \quad \lambda_1(B) \leq \frac{\int_B |\nabla u^*|^2}{\int_B (u^*)^2} \leq \frac{\int_\Omega |\nabla u|^2}{\int_\Omega u^2} = \lambda_1(\Omega),$$

the last equality because u is the first eigenfunction of Ω .

The decisive design choice is which geometric quantity of the superlevel sets the ball B is built to match. Schwarz symmetrisation matches *volume* and yields the Faber–Krahn inequality directly; the Kohler–Jobin rearrangement instead matches a *torsional functional*, the *modified torsional rigidity* introduced next, and this is exactly what allows the Dirichlet energy to be preserved (not merely decreased) while the L^2 norm grows. The relation $\lambda_1(B) \leq \lambda_1(\Omega)$ at equal *torsion* of B and Ω is then the inequality (??).

Standing regularity facts (cited, not proved).

- (R1) (*Regularity of the eigenfunction.*) The first eigenfunction u of Ω may be taken positive, solves $-\Delta u = \lambda u$ in Ω with $\lambda = \lambda_1(\Omega)$, and satisfies $u \in C^1(\Omega) \cap L^\infty(\Omega)$ (Moser iteration for the L^∞ bound, De Giorgi–Nash for C^1 ; see [?, Prop. 2.4]). Put $M := \|u\|_{L^\infty(\Omega)} < \infty$.
- (R2) (*Coarea formula.*) For $u \in W_{\text{loc}}^{1,1}$ and Borel $g \geq 0$, $\int_\Omega g |\nabla u| \, dx = \int_0^M (\int_{\{u=t\}} g \, d\mathcal{H}^{n-1}) \, dt$ (Federer; see Maggi, *Sets of Finite Perimeter*, Thm. 13.1).
- (R3) (*Saint–Venant inequality.*) For any open bounded $A \subset \mathbb{R}^n$, $T(A) \leq T(A^\#)$ where $A^\#$ is the ball with $|A^\#| = |A|$; in the scale-free form we use it,

$$(6) \quad T(A) \leq T_n \left(\frac{|A|}{\omega_n} \right)^{(n+2)/n},$$

with equality iff A is a ball. ([?, (1.7) with $p = 2$].)

- (R4) (*Distribution function.*) The distribution function $\mu_u(t) := |\{u > t\}|$ is non-increasing; together with the coarea formula this gives, for a.e. regular value t , that $\int_{\{u=t\}} |\nabla u|^{-1} \, d\mathcal{H}^{n-1}$ is finite and $\mu'_u(t) = - \int_{\{u=t\}} |\nabla u|^{-1} \, d\mathcal{H}^{n-1}$ (Brothers–Ziemer).

Two preliminary identities. We record the variational form of T and the level-set flux identity for the eigenfunction; both are elementary.

Lemma 1.2 (Variational torsion and the eigenfunction flux).

- (i) For open A of finite measure, with $F(w) := \int_A w \, dx - \frac{1}{2} \int_A |\nabla w|^2 \, dx$,

$$(7) \quad T(A) = 2 \max_{w \in H_0^1(A)} F(w),$$

the maximiser being the torsion function u_A ; equivalently $T(A) = \max\{(\int_A w)^2 / \int_A |\nabla w|^2 : 0 \neq w \in H_0^1(A)\}$.

- (ii) The first eigenfunction u satisfies, for a.e. $t \in (0, M)$,

$$(8) \quad \int_{\{u=t\}} |\nabla u| \, d\mathcal{H}^{n-1} = \lambda \int_{\{u>t\}} u \, dx.$$

Proof. ???: F is strictly concave and coercive on $H_0^1(A)$; its Euler equation is $-\Delta w = 1$, so the maximiser is u_A and $F(u_A) = \int u_A - \frac{1}{2} \int |\nabla u_A|^2 = \int u_A - \frac{1}{2} \int u_A = \frac{1}{2} \int u_A = \frac{1}{2} T(A)$, using $\int |\nabla u_A|^2 = \int u_A$ (test the equation by u_A). The second form is the standard Rayleigh-type characterisation of T ([?, Prop. 2.2] with $p = 2$).

???: integrate $-\Delta u = \lambda u$ over the (a.e. smooth) superlevel set $\{u > t\}$ and use the divergence theorem with outward normal $\nu = -\nabla u / |\nabla u|$ on $\partial\{u > t\} = \{u = t\}$: $\int_{\{u=t\}} |\nabla u| \, d\mathcal{H}^{n-1} = - \int_{\{u=t\}} \partial_\nu u \, d\mathcal{H}^{n-1} = \int_{\{u>t\}} (-\Delta u) \, dx = \lambda \int_{\{u>t\}} u \, dx$. $\square$

The modified torsional rigidity. The engine of the technique is the following functional, which assigns to a nonnegative function a torsion-type quantity that is sensitive to the *level structure* of the function, not just to its domain.

Definition 1.3 (Modified torsional rigidity). Let $u \in H_0^1(\Omega) \cap L^\infty(\Omega)$, $u \geq 0$, $M = \|u\|_\infty$, with distribution function $\mu(t) = |\{u > t\}|$. Assume the integrability condition

$$(9) \quad t \mapsto \frac{\mu(t)}{\int_{\{u=t\}} |\nabla u| d\mathcal{H}^{n-1}} \in L^\infty(0, M)$$

(satisfied by torsion functions and by first eigenfunctions, by (??); cf. [?, Lem. 3.2]). With $\mathcal{L} := \{g \in \text{Lip}([0, M]) : g(0) = 0\}$, define the *modified torsional rigidity of Ω along u*

$$(10) \quad T_{\text{mod}}(\Omega; u) := 2 \sup_{g \in \mathcal{L}} F(g \circ u), \quad F(w) = \int_{\Omega} w dx - \frac{1}{2} \int_{\Omega} |\nabla w|^2 dx.$$

Since the competitors $g \circ u$ form a subclass of $H_0^1(\Omega)$, comparison with (??) gives $T_{\text{mod}}(\Omega; u) \leq T(\Omega)$ always.

Lemma 1.4 (Closed form of T_{mod}). *Under the hypotheses of Definition ??, writing $P(t) := \int_{\{u=t\}} |\nabla u| d\mathcal{H}^{n-1}$, the supremum in (??) is uniquely attained by the nondecreasing g_0 with $g_0'(t) = \mu(t)/P(t)$, and*

$$(11) \quad T_{\text{mod}}(\Omega; u) = \int_0^M \frac{\mu(t)^2}{P(t)} dt.$$

Proof. We may restrict to nondecreasing g : replacing g by $\tilde{g}(t) = \int_0^t |g'|$ leaves $\int_{\Omega} |\nabla(g \circ u)|^2$ unchanged (since $\tilde{g}' = |g'|$ a.e.) and increases $\int_{\Omega} g \circ u$ (since $\tilde{g} \geq g$), hence increases F . For nondecreasing $g \in \mathcal{L}$, Cavalieri's principle gives $\int_{\Omega} g(u) dx = \int_0^M g'(t) \mu(t) dt$, while the coarea formula ?? gives $\int_{\Omega} |\nabla(g \circ u)|^2 dx = \int_0^M g'(t)^2 P(t) dt$. Thus

$$F(g \circ u) = \int_0^M \left[g'(t) \mu(t) - \frac{1}{2} g'(t)^2 P(t) \right] dt \leq \int_0^M \max_{s \geq 0} \left[s \mu(t) - \frac{1}{2} s^2 P(t) \right] dt = \int_0^M \frac{\mu(t)^2}{2P(t)} dt,$$

the pointwise maximum $\max_s (s\mu - \frac{1}{2}s^2P) = \mu^2/(2P)$ attained at $s = \mu(t)/P(t)$. Equality holds iff $g'(t) = \mu(t)/P(t)$ a.e., which by (??) is an L^∞ function, so $g_0 \in \mathcal{L}$. Multiplying by 2 gives (??). $\square$

The next proposition is the isoperimetric statement that makes the rearrangement decrease the Rayleigh quotient. It is the only place where the Saint–Venant inequality enters.

Proposition 1.5 (Isoperimetric inequality for T_{mod}). *Let $u \in H_0^1(\Omega) \cap L^\infty$, $u \geq 0$, satisfy (??). If $B \subset \mathbb{R}^n$ is a ball with $T(B) = T_{\text{mod}}(\Omega; u)$, then $|B| \leq |\Omega|$, with equality iff Ω is a ball and u is (a translate of) a radial function.*

Proof. $T(B) = T_{\text{mod}}(\Omega; u) \leq T(\Omega)$ by the remark after Definition ??. The Saint–Venant inequality (??) gives $T(\Omega) \leq T_n(|\Omega|/\omega_n)^{(n+2)/n}$, while $T(B) = T_n(|B|/\omega_n)^{(n+2)/n}$ by (??). Hence $(|B|/\omega_n)^{(n+2)/n} \leq (|\Omega|/\omega_n)^{(n+2)/n}$, i.e. $|B| \leq |\Omega|$. Equality forces $T_{\text{mod}}(\Omega; u) = T(\Omega)$ and equality in Saint–Venant, hence Ω a ball; the maximiser $g_0 \circ u$ of (??) then coincides with the (radial) torsion function, so u is radial. ([?, Prop. 3.8].) $\square$

The Kohler–Jobin rearrangement. We now build u^* . For $t \in [0, M]$ set $\Omega_t = \{u > t\}$ and let

$$(12) \quad \mathcal{T}(t) := T_{\text{mod}}(\Omega_t; (u - t)_+).$$

Since $|\{(u - t)_+ > s\}| = \mu(t + s)$ and the coarea factor of $(u - t)_+$ at level s is $P(t + s)$, formula (??) gives the explicit expression

$$(13) \quad \mathcal{T}(t) = \int_t^M \frac{\mu(\tau)^2}{P(\tau)} d\tau,$$

so that $\mathcal{T} \in \text{Lip}_{\text{loc}}([0, M])$, $\mathcal{T}(M) = 0$, $\mathcal{T}(0) = T_{\text{mod}}(\Omega; u) =: T_*$, and for a.e. t

$$(14) \quad \mathcal{T}'(t) = -\frac{\mu(t)^2}{P(t)} < 0.$$

Thus $\mathcal{T}$ is a strictly decreasing bijection $[0, M] \rightarrow [0, T_*]$.

Definition of u^* . Let $B = B_{R_*}$ be the ball with $T(B) = T_*$, i.e. $R_* = (T_*/T_n)^{1/(n+2)}$. For $\tau \in [0, T_*]$ let $R(\tau)$ be the radius with $T(B_{R(\tau)}) = \tau$, i.e.

$$(15) \quad R(\tau) = (\tau/T_n)^{1/(n+2)}, \quad |B_{R(\tau)}| = \omega_n R(\tau)^n = \omega_n (\tau/T_n)^{n/(n+2)}.$$

Let $\varphi := \mathcal{T}^{-1} : [0, T_*] \rightarrow [0, M]$ (so $\varphi(0) = M$, $\varphi(T_*) = 0$, $\varphi' < 0$). We define u^* on B to be the radial decreasing function whose value on the sphere $\{|x| = R(\tau)\}$ is $\psi(\tau)$, where $\psi : [0, T_*] \rightarrow [0, \infty)$ is the Lipschitz function determined by

$$(16) \quad (-\psi'(\tau)) \mu^*(\psi(\tau)) = (-\varphi'(\tau)) \mu(\varphi(\tau)), \quad \psi(T_*) = 0,$$

μ^* denoting the distribution function of u^* . Concretely the superlevel set $\{u^* > \psi(\tau)\} = B_{R(\tau)}$ has, by construction, torsional rigidity exactly τ , the value of the *modified* torsion that the corresponding superlevel set $\Omega_{\varphi(\tau)}$ of u carries. In particular $\{u^* > 0\} = B = B_{R_*}$, so

$$(17) \quad T(B) = T_* = T_{\text{mod}}(\Omega; u) \leq T(\Omega).$$

Since $\mu^*(\psi(\tau)) = |B_{R(\tau)}| = \omega_n R(\tau)^n$, equation (??) integrates to an explicit ψ ; we shall only need the two qualitative consequences ??, ??, proved next.

Lemma 1.6 (Dirichlet energy is preserved). $\int_B |\nabla u^*|^2 dx = \int_\Omega |\nabla u|^2 dx$.

Proof. The original side. By coarea and the change of variable $t = \varphi(\tau)$ (Lemma 2.5 of [?], the one-dimensional area formula for $\varphi = \mathcal{T}^{-1} \in \text{Lip}_{\text{loc}}$ with $\varphi' < 0$),

$$(18) \quad \int_\Omega |\nabla u|^2 dx = \int_0^M P(t) dt = \int_0^{T_*} P(\varphi(\tau)) (-\varphi'(\tau)) d\tau.$$

Differentiating $\mathcal{T}(\varphi(\tau)) = \tau$ and using (??) gives the key relation

$$(19) \quad -\varphi'(\tau) = \frac{1}{-\mathcal{T}'(\varphi(\tau))} = \frac{P(\varphi(\tau))}{\mu(\varphi(\tau))^2}.$$

Hence, writing $P(\varphi)(-\varphi') = \mu(\varphi)^2 \cdot \frac{P(\varphi)}{\mu(\varphi)^2} \cdot (-\varphi')$ and using (??) for the middle factor, $P(\varphi)(-\varphi') = \mu(\varphi)^2(-\varphi')^2$, so (??) becomes

$$(20) \quad \int_\Omega |\nabla u|^2 dx = \int_0^{T_*} \mu(\varphi(\tau))^2 (-\varphi'(\tau))^2 d\tau.$$

The rearranged side. For the radial u^* , the level set $\{u^* = \psi(\tau)\}$ is the sphere of radius $R(\tau)$, on which $|\nabla u^*|$ is constant. A direct computation (differentiate $u^*(x) = \psi(\tau)$ on $|x| = R(\tau)$ and use $R(\tau) = (\tau/T_n)^{1/(n+2)}$) gives, with $P^*(\tau') := \int_{\{u^* = \tau'\}} |\nabla u^*| d\mathcal{H}^{n-1}$ the coarea factor of u^* ,

$$(21) \quad \int_B |\nabla u^*|^2 dx = \int_0^{T_*} \mu^*(\psi(\tau))^2 (-\psi'(\tau))^2 d\tau,$$

the exact radial analogue of (??); this is the specialisation to $p = 2$ of the identity (4.4) of [?], obtained there by the same coarea computation. Now the defining relation (??), $\mu^*(\psi)(-\psi') = \mu(\varphi)(-\varphi')$, makes the integrands of (??) and (??) equal:

$$\mu^*(\psi(\tau))^2 (-\psi'(\tau))^2 = [\mu^*(\psi(\tau))(-\psi'(\tau))]^2 = [\mu(\varphi(\tau))(-\varphi'(\tau))]^2 = \mu(\varphi(\tau))^2 (-\varphi'(\tau))^2.$$

Hence $\int_B |\nabla u^*|^2 = \int_\Omega |\nabla u|^2$. $\square$

Lemma 1.7 (L^2 norm does not decrease). $\int_B (u^*)^2 dx \geq \int_\Omega u^2 dx$, and more generally $\int_B f(u^*) dx \geq \int_\Omega f(u) dx$ for every convex $f : [0, \infty) \rightarrow [0, \infty)$ with $f(0) = 0$, strictly if f is strictly convex unless Ω is a ball and u radial.

Proof. The point is the comparison of *distribution functions* along the rearrangement. Apply Proposition ?? not to u but to the truncations: for each $\tau \in [0, T_\star]$ the ball $B_{R(\tau)}$ has $T(B_{R(\tau)}) = \tau = T_{\text{mod}}(\Omega_{\varphi(\tau)}; (u - \varphi(\tau))_+)$ by construction, so Proposition ?? (applied to $\Omega_{\varphi(\tau)}$ with reference function $(u - \varphi(\tau))_+$) yields

$$(22) \quad \mu^*(\psi(\tau)) = |B_{R(\tau)}| \leq |\Omega_{\varphi(\tau)}| = \mu(\varphi(\tau)), \quad \tau \in [0, T_\star].$$

Combining (??) with the defining relation (??), $\mu^*(\psi)(-\psi') = \mu(\varphi)(-\varphi')$, gives

$$(23) \quad -\psi'(\tau) = \frac{\mu(\varphi(\tau))}{\mu^*(\psi(\tau))} (-\varphi'(\tau)) \geq -\varphi'(\tau) \quad \text{for a.e. } \tau,$$

since $\mu(\varphi) \geq \mu^*(\psi)$. Integrating from τ to $T_\star$ and using $\psi(T_\star) = \varphi(T_\star) = 0$,

$$(24) \quad \psi(\tau) = \int_\tau^{T_\star} (-\psi'(s)) ds \geq \int_\tau^{T_\star} (-\varphi'(s)) ds = \varphi(\tau), \quad \tau \in [0, T_\star].$$

Now compute, for convex f with $f(0) = 0$ (so $f(u) = \int_0^u f'(t) dt$ and Cavalieri applies), using the change of variable $t = \varphi(\tau)$:

$$\begin{aligned} \int_\Omega f(u) dx &= \int_0^M f'(t) \mu(t) dt = \int_0^{T_\star} f'(\varphi(\tau)) (-\varphi'(\tau)) \mu(\varphi(\tau)) d\tau \\ &= \int_0^{T_\star} f'(\varphi(\tau)) (-\psi'(\tau)) \mu^*(\psi(\tau)) d\tau \quad \text{by (??)} \\ &\leq \int_0^{T_\star} f'(\psi(\tau)) (-\psi'(\tau)) \mu^*(\psi(\tau)) d\tau \quad \text{by (??) and } f' \text{ nondecreasing} \\ &= \int_0^M f'(s) \mu^*(s) ds = \int_B f(u^*) dx, \end{aligned}$$

the last change of variable being $s = \psi(\tau)$. Taking $f(s) = s^2$ proves $\int_B (u^*)^2 \geq \int_\Omega u^2$. If f is strictly convex, equality in the middle step forces $\psi = \varphi$ a.e., hence equality in (??), hence Ω a ball and u radial by the equality case of Proposition ??.

Proof of Theorem ??. Apply the construction to the first eigenfunction u of Ω (a valid reference function by (??) and ??). By Lemmas ?? and ?? the rearranged $u^* \in H_0^1(B)$ satisfies $\int_B |\nabla u^*|^2 = \int_\Omega |\nabla u|^2$ and $\int_B (u^*)^2 \geq \int_\Omega u^2$. By the Rayleigh principle on B (??) and these two facts,

$$\lambda_1(B) \leq \frac{\int_B |\nabla u^*|^2}{\int_B (u^*)^2} \leq \frac{\int_\Omega |\nabla u|^2}{\int_\Omega u^2} = \lambda_1(\Omega).$$

Multiplying $\lambda_1(B) \leq \lambda_1(\Omega)$ by $T(B)^\beta > 0$ and using $T(B) \leq T(\Omega)$ from (??) and the scale invariance $\Psi(B) = \Psi(B_1)$:

$$T(B_1) L_n^\beta = \Psi(B) = T(B) \lambda_1(B)^\beta \leq T(\Omega) \lambda_1(\Omega)^\beta = \Psi(\Omega).$$

This is (??) (for an arbitrary ball, $\Psi(B) = \Psi(B_1)$ on both sides). Equality forces equality throughout, in particular in Lemma ?? with the strictly convex $f(s) = s^2$, hence Ω is a ball. $\square$

Remark 1.8 (On the sign that distinguishes Kohler–Jobin from Faber–Krahn). The torsion-preserving rearrangement makes the ball $B_{R(\tau)}$ *smaller* than the superlevel set $\Omega_{\varphi(\tau)}$, (??) — the opposite of Schwarz symmetrisation. It is precisely this that lets the Dirichlet energy be *exactly preserved* (Lemma ??) rather than merely decreased, while the level-set volumes *grow under the inverse rearrangement*, forcing $\int (u^*)^2 \geq \int u^2$ (Lemma ??). The naive attempt to compare L^2 masses level by level fails because the rearranged superlevel balls are smaller; the correct comparison is through the *distribution-function inequality* (??), driven by the isoperimetric Proposition ??.

1.3. Part 2: transfer to sharp Faber–Krahn stability. We now use Theorem ?? as a tool. Throughout, $L := \lambda_1(B^*)$, $T_0 := T(B^*)$, $\beta = (n+2)/2$, $p := 1/\beta = 2/(n+2) \in (0, 1]$.

Lemma 1.9 (Pointwise Kohler–Jobin transfer). *For every open $\Omega \subset \mathbb{R}^n$ with $|\Omega| = |B^*|$,*

$$(25) \quad \lambda_1(\Omega) \geq L \left(\frac{T_0}{T(\Omega)} \right)^{1/\beta} = L \left(\frac{T_0}{T(\Omega)} \right)^{2/(n+2)}.$$

Proof. Theorem ?? with $B = B^*$ gives $T(\Omega)\lambda_1(\Omega)^\beta \geq T_0 L^\beta$; solve for $\lambda_1(\Omega)$. $\square$

By Saint–Venant (??) at fixed volume, $T(\Omega) \leq T_0$, so the torsion deficit $\sigma := \sigma(\Omega) := T_0 - T(\Omega) \geq 0$. Put $s := \sigma/T_0 \in [0, 1]$; then (??) reads

$$(26) \quad \lambda_1(\Omega) - L \geq L[(1-s)^{-p} - 1], \quad p = \frac{2}{n+2}.$$

Lemma 1.10 (An elementary one-variable inequality). *For $p \in (0, 1]$ and $s \in [0, 1]$, $(1-s)^{-p} - 1 \geq ps$.*

Proof. $f(s) = (1-s)^{-p} - 1$ has $f(0) = 0$ and $f'(s) = p(1-s)^{-(p+1)} \geq p$ on $[0, 1]$ because $(1-s)^{-(p+1)} \geq 1$. By the fundamental theorem of calculus $f(s) = \int_0^s f' \geq ps$. $\square$

Lemma 1.11 (Small-deficit linearisation). *If $|\Omega| = |B^*|$ and $\sigma \leq T_0/2$, then $\lambda_1(\Omega) - L \geq \frac{2L}{(n+2)T_0} \sigma$.*

Proof. With $s = \sigma/T_0 \leq 1/2$, $p = 2/(n+2)$, combine (??) and Lemma ??: $\lambda_1(\Omega) - L \geq Lps = Lp\sigma/T_0 = \frac{2L}{(n+2)T_0} \sigma$. $\square$

Lemma 1.12 (Large-deficit branch). *If $|\Omega| = |B^*|$ and $\sigma > T_0/2$, then $\lambda_1(\Omega) - L \geq L(2^{2/(n+2)} - 1)$, hence, since $\mathcal{A}(\Omega) < 2$,*

$$(27) \quad \lambda_1(\Omega) - L \geq \frac{L(2^{2/(n+2)} - 1)}{4} \mathcal{A}(\Omega)^2.$$

Proof. $\sigma > T_0/2$ gives $T(\Omega) < T_0/2$, so $T_0/T(\Omega) > 2$ and (??) yields $\lambda_1(\Omega) \geq L 2^{2/(n+2)}$. Since $\mathcal{A}(\Omega)^2 \leq 4$ at fixed volume, dividing by 4 gives (??). $\square$

Theorem 1.13 (Sharp Faber–Krahn via Kohler–Jobin). *Assume the global sharp Saint–Venant stability estimate: there is $c_{\text{SV}}^{\text{glob}}(n) > 0$ such that*

$$(28) \quad T(B^*) - T(\Omega) = 2(E(\Omega) - E(B^*)) \geq 2c_{\text{SV}}^{\text{glob}}(n) \left(\frac{|\Omega|}{\omega_n} \right)^{(n+2)/n} \mathcal{A}(\Omega)^2$$

for all open bounded $\Omega \subset \mathbb{R}^n$, $0 < |\Omega| < \infty$. Then

$$(29) \quad \delta_{\text{FK}}(\Omega) = \left(\frac{|\Omega|}{\omega_n} \right)^{2/n} (\lambda_1(\Omega) - \lambda_1(B^*)) \geq c_{\text{FK}}(n) \mathcal{A}(\Omega)^2,$$

with the explicit constant

$$(30) \quad c_{\text{FK}}(n) = \min \left\{ \frac{4L_n c_{\text{SV}}^{\text{glob}}(n)}{(n+2)T_n}, \frac{L_n(2^{2/(n+2)} - 1)}{4} \right\},$$

equivalently $\mathcal{A}(\Omega)^2 \leq C_{\text{FK}}(n) \delta_{\text{FK}}(\Omega)$ with $C_{\text{FK}}(n) = 1/c_{\text{FK}}(n)$.

Proof. Both δ_{FK} and $\mathcal{A}$ are scale invariant, and so is (??); rescale so that $|\Omega| = \omega_n$, whence $B^* = B_1$, $L = L_n$, $T_0 = T_n$, and the volume prefactor $(|\Omega|/\omega_n)^{2/n} = 1$. Split on the torsion deficit $\sigma = T_n - T(\Omega)$.

Case 1 ($\sigma \leq T_n/2$). By Lemma ?? and (??) (which now reads $\sigma \geq 2c_{\text{SV}}^{\text{glob}}(n)\mathcal{A}(\Omega)^2$),

$$\lambda_1(\Omega) - L_n \geq \frac{2L_n}{(n+2)T_n} \sigma \geq \frac{2L_n}{(n+2)T_n} \cdot 2c_{\text{SV}}^{\text{glob}}(n) \mathcal{A}(\Omega)^2 = \frac{4L_n c_{\text{SV}}^{\text{glob}}(n)}{(n+2)T_n} \mathcal{A}(\Omega)^2.$$

Case 2 ($\sigma > T_n/2$). By Lemma ??, $\lambda_1(\Omega) - L_n \geq \frac{L_n(2^{2/(n+2)} - 1)}{4} \mathcal{A}(\Omega)^2$.

In both cases $\delta_{\text{FK}}(\Omega) = \lambda_1(\Omega) - L_n \geq c_{\text{FK}}(n) \mathcal{A}(\Omega)^2$ with $c_{\text{FK}}(n)$ the minimum (??). Undoing the scaling preserves the inequality since every term is scale invariant. $\square$

Remark 1.14 (Sharpness of the exponent). The exponent 2 on $\mathcal{A}$ in (??) is optimal: smooth ellipsoidal perturbations of the ball have $\lambda_1(\Omega) - L_n \asymp \mathcal{A}(\Omega)^2$ ([?, ?] and references therein). The transfer is non-perturbative: no compactness, contradiction, or Taylor expansion is used past the elementary Lemma ??.

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